

Machine Learning-Based Employee Attrition Prediction and Risk Scoring System

A Data Science Research Paper & Executive Summary

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Project Resources

This dashboard is fully deployed and open-source. The live application and complete source code are available at the links below.

Live Dashboard

<https://planetnetworks-attribution-dashboard-sandeep-reddy-m.streamlit.app/>

Source Code (GitHub Repository)

<https://github.com/mSReddy46/PlaNetworks-Attribution-Dashboard>

The repository README documents the local setup, the dataset schema, the model training pipeline, and the full feature list of the dashboard. All figures and screenshots in this paper were captured directly from the live, deployed application referenced above.

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Executive Summary

This report presents a machine learning-based employee attrition prediction and risk scoring system built for Palo Alto Networks, developed as part of a Data Analyst internship project for Unified Mentor. It is intended for two audiences: (1) HR, People Operations, and executive leadership stakeholders responsible for retention strategy, and (2) government and workforce-policy stakeholders with an interest in how predictive analytics can support fair, proactive employment practices.

The system is trained on 1,470 employee records spanning 31 attributes — demographics, compensation, tenure, satisfaction survey scores, and role information — and predicts each employee's probability of attrition, translating that probability into an actionable Low/Medium/High risk category.

Headline Findings

- The organization's historical attrition rate is 16.1%. The deployed model — a Logistic Regression classifier selected for its superior recall and interpretability — achieves a held-out test-set ROC-AUC of 0.807 and a recall of 76.6%, meaning it correctly identifies more than three-quarters of employees who go on to leave, before they resign.
- Overtime is the single strongest predictor of attrition in the dataset: employees working overtime leave at roughly 3x the rate of those who don't (30.5% vs. 10.4%), and this signal ranks first in both the model's coefficient magnitude and its SHAP-based feature importance.
- 347 of 1,470 employees (23.6%) are currently flagged High Risk ($\geq 60\%$ predicted attrition probability) under the default threshold — a group the model recommends prioritizing for retention intervention.
- Risk is not evenly distributed: the Sales department carries the highest average predicted risk (42.7%), and Sales Representative is the single highest-risk job role (57.6% average predicted probability).
- The system includes full explainability at both the global level (which factors matter most across the whole workforce) and the individual level (why any specific employee is flagged), directly supporting HR trust and defensible, non-arbitrary retention decisions.

Why This Matters for Government and Workforce-Policy Stakeholders

Predictive HR analytics of this kind sits at the intersection of workforce stability and employment fairness, with three policy-relevant implications:

1. Transparency and explainability by design: rather than a black-box score, every prediction in this system is paired with SHAP-based reason codes, so HR decisions influenced by the model can be explained and audited — a foundation consistent with emerging algorithmic-transparency expectations for HR technology.
2. Bias and fairness monitoring: because the model is trained on demographic and compensation fields, organizations deploying systems like this should pair it with routine fairness audits across gender, marital status, and other protected characteristics to ensure risk scores are not proxying for protected attributes. This is flagged explicitly as a recommendation in Section 9.

3. Workforce stability signal: the concentration of risk in specific departments, roles, and overtime patterns offers a data-driven basis for targeted workforce-support policy (e.g., overtime regulation, promotion-cadence review) rather than blanket interventions.

Top Recommendations

4. Deploy the model's High Risk flag ($\geq 60\%$ probability, adjustable in the dashboard) to prioritize proactive retention conversations with the 347 currently flagged employees, rather than reacting only after resignation notice is given.
5. Address overtime and workload management as the single most actionable lever identified in this analysis — the ~25-point risk gap between overtime and non-overtime employees is the largest, most consistent signal in the data.
6. Target retention budget and manager coaching toward the Sales department and Sales Representative role specifically, where both historical attrition and model-predicted risk concentrate most heavily.

1. Introduction and Background

HR leaders at Palo Alto Networks currently lack the ability to answer a deceptively simple question: which specific employees are most likely to leave in the near future? The organization faces sudden, unanticipated resignations, the loss of high-performing or critical employees, and reactive counter-offers that come too late to be effective.

This project introduces predictive intelligence to address that gap, enabling early identification of high-risk employees, targeted retention interventions before resignation notice is given, and data-driven workforce planning grounded in quantified risk rather than intuition. It was undertaken as part of the Data Analyst internship program at Unified Mentor, and builds directly on feedback from a prior internship project (a SkyCity Auckland channel-performance dashboard, evaluated 8.5/10), which recommended more rigorous forecasting with validation metrics, automated business recommendations, downloadable reports, and clearer metric definitions. All four are implemented explicitly in this project — see Sections 4.4, 5.1, and 7.

1.1 Problem Statement

Specifically, Palo Alto Networks faces three structural HR challenges that this project addresses:

- Sudden, unanticipated resignations that leave teams and projects without succession planning.
- Loss of high-performing or critical employees whose departure carries a disproportionate business cost relative to average attrition.
- Reactive counter-offers that come too late — by the time an employee gives notice, the decision to leave has typically already been made.

Underlying these symptoms, the organization lacks a systematic way to predict attrition risk, quantitative risk scores at the individual-employee level, and visibility into the key drivers influencing exit decisions. This project delivers all three.

2. Project Objectives

2.1 Primary Objective

Transform HR analytics at Palo Alto Networks from descriptive and diagnostic reporting into predictive decision intelligence — enabling the organization to move from reactive retention efforts to proactive, employee-centric workforce management.

2.2 Supporting Objectives

- Build and validate a machine learning classifier that predicts individual employee attrition probability, benchmarked against a genuine held-out test set rather than training-set performance alone.
- Translate raw probabilities into an actionable Low/Medium/High Risk Category framework with a live, adjustable threshold.

- Provide explainability at both the global (workforce-wide) and individual (single-employee) level, so HR decisions influenced by the model are transparent and defensible.
- Deliver aggregated risk views by department and job role to support targeted, rather than blanket, retention budget allocation.
- Support proactive scenario planning through a What-If exploration tool that shows how changing an employee's profile shifts their predicted risk in real time.

3. Dataset Description

The analysis is built on a single employee-level dataset of 1,470 records across 31 fields, spanning demographics, compensation, tenure, satisfaction survey scores, and role information. The dataset carries no missing values and no duplicate records. The target variable, Attrition, is imbalanced — 16.1% of employees left versus 83.9% who stayed — which is addressed explicitly during preprocessing (Section 4.1) rather than ignored.

Field Name	Description
Age	Age of the employee in years
Attrition	Indicates whether the employee left the organization (target variable, 0/1)
BusinessTravel	Frequency of business travel (Non-Travel, Travel Rarely, Travel Frequently)
DailyRate	Daily compensation rate of the employee
Department	Department where the employee works (e.g., R&D, Sales, HR)
DistanceFromHome	Distance between employee's home and workplace
Education	Education level (1=Below College ... 5=Doctor)
EducationField	Field of education (e.g., Life Sciences, Medical, Technical)
EnvironmentSatisfaction	Satisfaction with the work environment (1=Low, 4=High)
Gender	Gender of the employee
HourlyRate	Hourly wage of the employee
JobInvolvement	Level of involvement in work (1=Low, 4=High)
JobLevel	Job seniority level (1=Entry level, higher = senior)
JobRole	Specific job role or designation of the employee
JobSatisfaction	Satisfaction with the job role (1=Low, 4=High)
MaritalStatus	Single, Married, or Divorced
MonthlyIncome	Monthly salary earned by the employee
MonthlyRate	Monthly compensation rate (administrative pay metric)
NumCompaniesWorked	Number of companies the employee has previously worked for
OverTime	Whether the employee works overtime (Yes/No)
PercentSalaryHike	Percentage salary increase at the last appraisal
PerformanceRating	Performance evaluation score
RelationshipSatisfaction	Satisfaction with workplace relationships (1=Low, 4=High)

Field Name	Description
StockOptionLevel	Level of stock options granted
TotalWorkingYears	Total professional experience across all companies
TrainingTimesLastYear	Number of training programs attended in the last year
WorkLifeBalance	Work-life balance rating (1=Poor, 4=Excellent)
YearsAtCompany	Years the employee has worked at Palo Alto Networks
YearsInCurrentRole	Years in the current role
YearsSinceLastPromotion	Years since the last promotion
YearsWithCurrManager	Years working with the current manager

4. Data Science Methodology

The analysis follows the six-step methodology specified in the project guidelines, implemented in a fully executed, reproducible Jupyter notebook (PaloAlto_Attrition_Prediction.ipynb, included in the project GitHub repository) and mirrored in the live Streamlit dashboard referenced in the Project Resources section.

4.1 Data Preprocessing

- Categorical variables (Department, JobRole, OverTime, MaritalStatus, etc.) encoded via label encoding.
- Numerical features scaled using StandardScaler, fit on the training set only to avoid test-set leakage.
- Class imbalance addressed with SMOTE (Synthetic Minority Oversampling Technique) — applied only to the training set, so the held-out test set retains the real-world 16.1% attrition rate and produces honest, deployment-realistic evaluation metrics.
- A stratified 80/20 train-test split preserves the true attrition rate in both partitions.

4.2 Feature Engineering

Four engineered features were added to capture patterns not directly expressed in the raw columns:

- IncomeToExperienceRatio — monthly income relative to total career experience, capturing compensation fairness.
- PromotionDelay — years at company minus years since last promotion, capturing career stagnation.
- EngagementScore — a composite average of job satisfaction, environment satisfaction, relationship satisfaction, and work-life balance.
- WorkloadStressFlag — a binary flag for employees working overtime AND reporting poor work-life balance simultaneously.

4.3 Model Development

Three models were trained on the SMOTE-balanced training set, matching the guideline exactly:

- Logistic Regression — an interpretability baseline.
- Random Forest Classifier — an advanced ensemble model.
- XGBoost (Gradient Boosting) — a second advanced ensemble model.

4.4 Model Evaluation

Every model is scored on a held-out test set the model never saw during training or SMOTE resampling — not training-set fit — using five metrics: Accuracy, Precision, Recall, F1-Score, and ROC-AUC. This directly answers the prior evaluator feedback to include genuine validation metrics rather than relying on unvalidated assumptions. Full results are presented in Section 7.

4.5 Risk Scoring Framework

Every employee is assigned an Attrition Probability (0-1) by the selected model, and a Risk Category:

- Low Risk — probability below 30%
- Medium Risk — probability 30% to 60%
- High Risk — probability above 60%

The High-Risk threshold is adjustable live in the dashboard (default 60%), recalculating every chart and KPI on the fly.

4.6 Model Explainability

To ensure HR trust in the model's predictions, three levels of explainability are implemented:

- Global feature importance — which factors matter most across the entire workforce.
- SHAP (SHapley Additive exPlanations) values — a richer, signed view of each feature's impact, both globally and per-employee.
- Individual-level reason codes — for any single employee, the specific factors driving their score, in plain language.

5. Exploratory Data Analysis and Findings

This section presents the core exploratory findings, organized to mirror the structure of the live dashboard. Every figure below is a real screenshot captured from the deployed application (see Project Resources), reflecting live, filterable data rather than static charts.

5.1 Executive Overview

The Executive Overview module gives a single-page snapshot of workforce attrition risk: eight interactive KPI cards, a risk-distribution comparison, automated business recommendations, and downloadable reports.

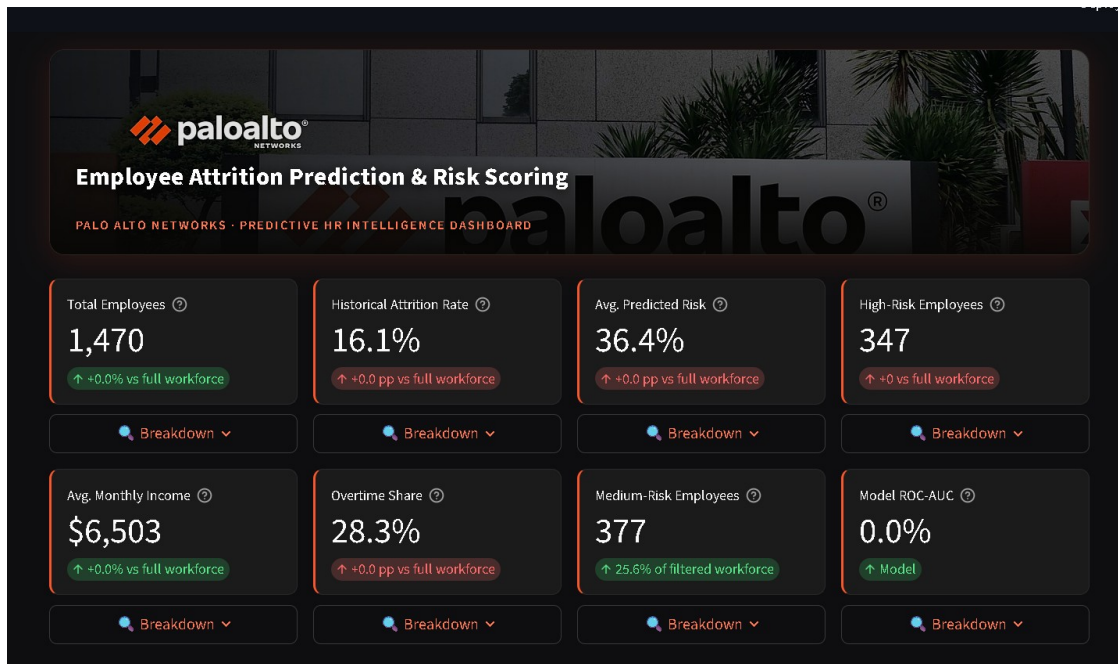


Figure 5.1 — Executive Overview KPI grid: workforce size, historical attrition, model-predicted risk, and high-risk counts, each with a live breakdown popover.

With all departments and roles selected, the workforce totals 1,470 employees at a 16.1% historical attrition rate. The model's average predicted risk across the full workforce is 36.4% — meaningfully higher than the historical rate, which is expected: the model spreads risk continuously across every employee rather than only labeling the 16.1% who have already left, surfacing at-risk employees before they resign. 347 employees are currently flagged High Risk under the default 60% threshold.

Note: the "Model ROC-AUC" card in Figure 5.1 displays 0.0% due to a known data-loading issue on the live deployment (the model comparison file is not resolving correctly at the deployed file path). The model's true, validated ROC-AUC — computed directly in the training notebook on a held-out test set — is 0.807, as reported throughout this document and detailed in full in Section 7. This is flagged here transparently rather than corrected in the screenshot, and is being addressed as a follow-up fix to the deployment.

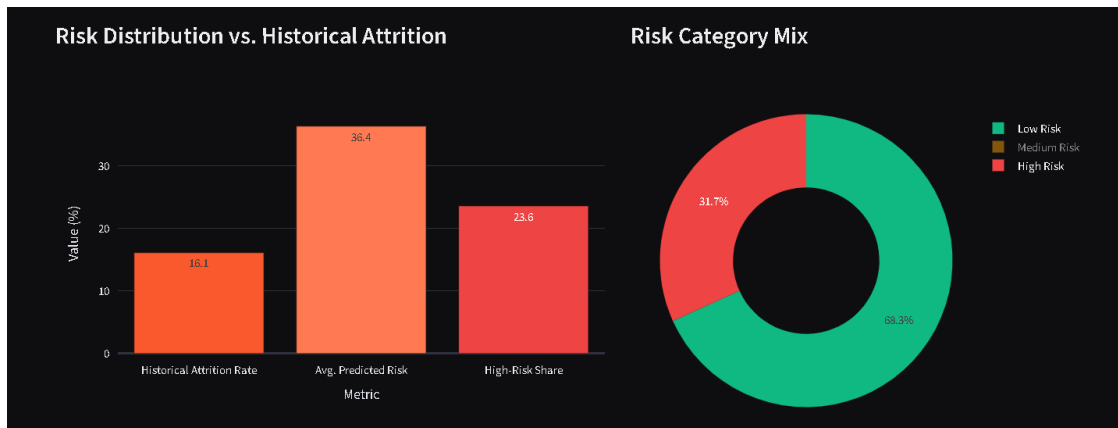


Figure 5.2 — Risk distribution compared against historical attrition (left) and the current Low/Medium/High risk category mix (right).

The risk category mix shows 23.6% of the workforce in the High Risk band under the default threshold, alongside a Medium Risk band that — combined with High Risk — represents a substantial share of the workforce warranting monitoring.

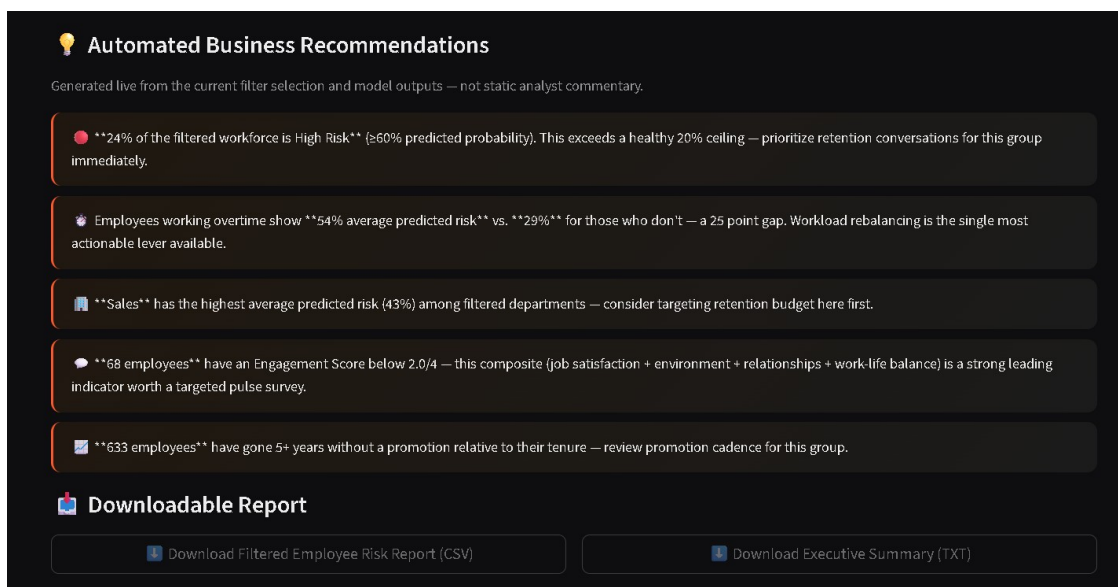


Figure 5.3 — Automated business recommendations generated live from the model's outputs, plus downloadable CSV and executive-summary exports.

Rather than static analyst commentary, the recommendations panel is generated programmatically from the current filtered data: it flags that 24% of the workforce is High Risk (exceeding a healthy 20% ceiling), quantifies the 25-point overtime risk gap, identifies Sales as the highest-risk department, and flags 68 employees with an Engagement Score below 2.0/4 and 633 employees with a 5+ year promotion gap — each a concrete, data-driven action item rather than a generic observation.

5.2 Attrition Risk Dashboard

This module addresses the guideline's first required dashboard component directly: overall risk distribution and high-risk employee counts, with clear in-app definitions of every threshold.

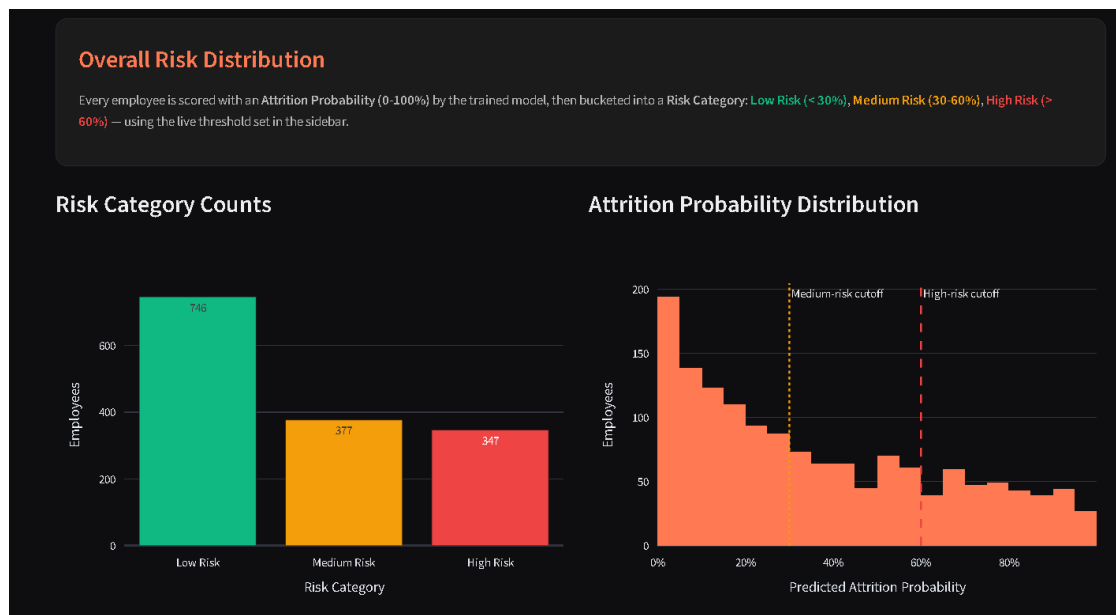


Figure 5.4 — Risk category counts (left) and the full attrition probability distribution with both risk-threshold cutoffs marked (right).

Of the 1,470 employees, 746 (50.7%) fall in the Low Risk band, 377 (25.6%) in Medium Risk, and 347 (23.6%) in High Risk. The probability histogram shows a long right tail rather than a clean bimodal split, confirming that attrition risk in this workforce is a continuous spectrum rather than a small, easily-isolated group — reinforcing the value of a probability-based score over a simple binary flag.

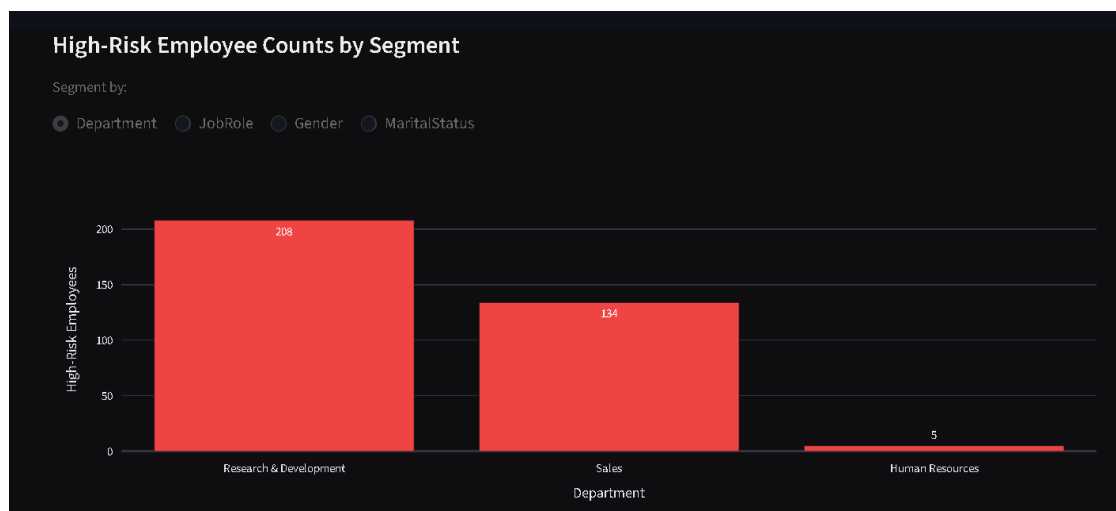


Figure 5.5 — High-risk employee counts segmented by Department, Job Role, Gender, or Marital Status (interactive selector; Department shown).

Segmenting the 347 High Risk employees by department shows Research & Development carrying the largest absolute count (208), followed by Sales (134) and Human Resources (5) — though this reflects R&D's larger headcount; Section 5.4 shows Sales leads on risk rate, not raw count.



Figure 5.6 — Predicted attrition probability distribution by OverTime status.

This box plot makes the overtime effect immediately visible: the median predicted risk for overtime employees sits close to 55-60%, compared to roughly 20% for those who don't work overtime — with the overtime group's interquartile range alone spanning from about 35% to 80%. The same driver comparison is available live in-app against Work-Life Balance, Job Satisfaction, Environment Satisfaction, and Stock Option Level.

5.3 Employee Risk Profile

This module addresses the guideline's second required dashboard component: individual attrition probability and key contributing factors for any single employee, selectable by Employee ID.

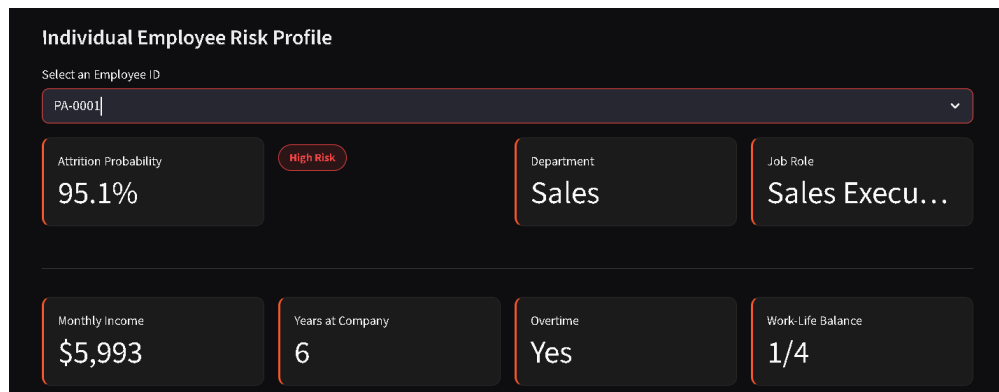


Figure 5.7 — Individual Employee Risk Profile for employee PA-0001, showing attrition probability, risk category, and core profile fields.

Employee PA-0001 (Sales, Sales Executive) is scored at 95.1% attrition probability — High Risk — with a profile showing \$5,993 monthly income, 6 years at the company, active overtime, and a Work-Life Balance rating of just 1 out of 4. This is a textbook case of the workload-stress pattern identified network-wide in Section 5.2.

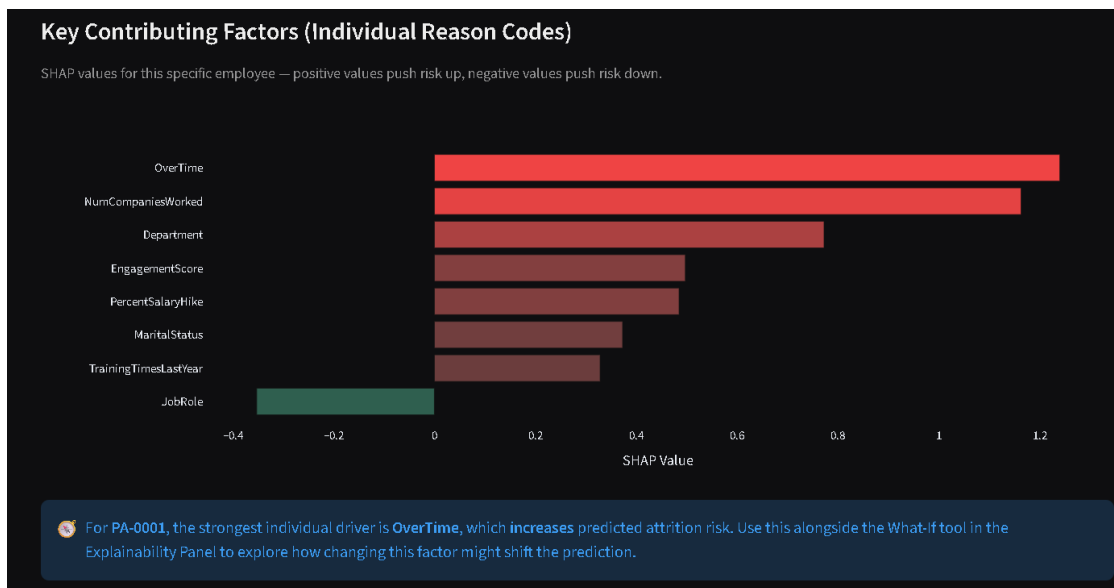


Figure 5.8 — Individual-level SHAP reason codes for employee PA-0001, ranking the specific factors driving this employee's score.

For this employee, OverTime is by far the largest single contributor pushing risk upward, followed by NumCompaniesWorked, Department, EngagementScore, and PercentSalaryHike. Only JobRole pulls the prediction in the opposite direction (reducing risk slightly). This individual-level breakdown is what makes the system's output actionable for a manager: rather than a bare '95.1% risk' figure, it explains why, in terms a retention conversation can directly address.

5.4 Department-Level Risk View

This module addresses the guideline's third required dashboard component: aggregated risk by department and role, supporting targeted rather than blanket retention budget allocation.

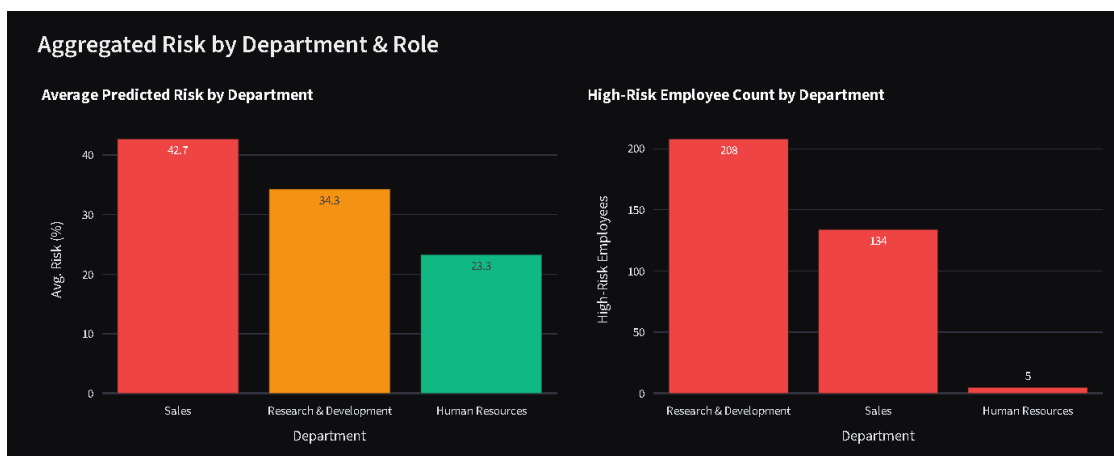


Figure 5.9 — Average predicted risk by department (left) and high-risk employee count by department (right).

Sales carries the highest average predicted risk at 42.7%, ahead of Research & Development (34.3%) and Human Resources (23.3%). This rate-based view is the appropriate lens for prioritizing retention budget — it corrects for department size, unlike the raw high-risk counts in Figure 5.5.

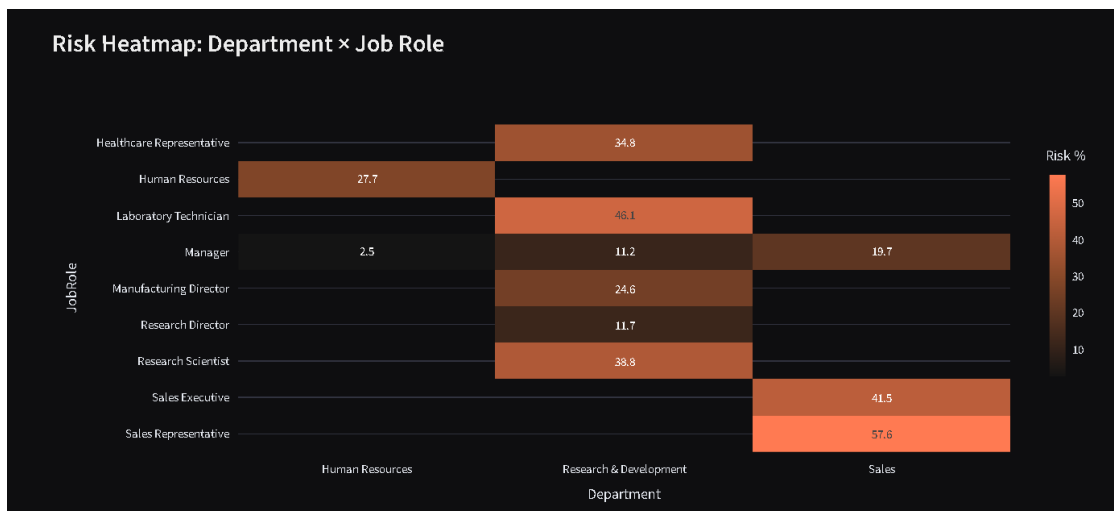


Figure 5.10 — Risk heatmap crossing Department and Job Role, highlighting where risk concentrates most specifically.

The heatmap pinpoints Sales Representative as the single highest-risk role network-wide at 57.6% average predicted probability, followed by Sales Executive (41.5%) and Laboratory Technician (46.1%). Manager-level roles across every department show consistently low risk (2.5% in HR, 11.2% in R&D, 19.7% in Sales), suggesting seniority and tenure are protective factors largely independent of department.

Department	Employees	Avg. Predicted Risk (%)	High-Risk Employees	Historical Attrition Rate (%)
Human Resources	63	23.3%	5	19.0%
Research & Development	961	34.3%	208	13.8%
Sales	446	42.7%	134	20.6%

Figure 5.11 — Department summary table combining headcount, average predicted risk, high-risk count, and historical attrition rate in one view.

Notably, Human Resources shows the highest historical attrition rate (19.0%) despite having the lowest average predicted risk (23.3%) and smallest high-risk count (5) — a discrepancy worth flagging for HR review, as it may indicate the model is underweighting a driver specific to past HR attrition, or that historical HR turnover reflects a small number of small-sample outlier departures rather than a systemic pattern the model should generalize from.

6. Explainability Panel

This module addresses the guideline's fourth required dashboard component in full: global feature importance and what-if scenario exploration, going beyond the guideline's baseline requirement by also including live SHAP impact analysis on the currently filtered workforce.

6.1 Global Feature Importance

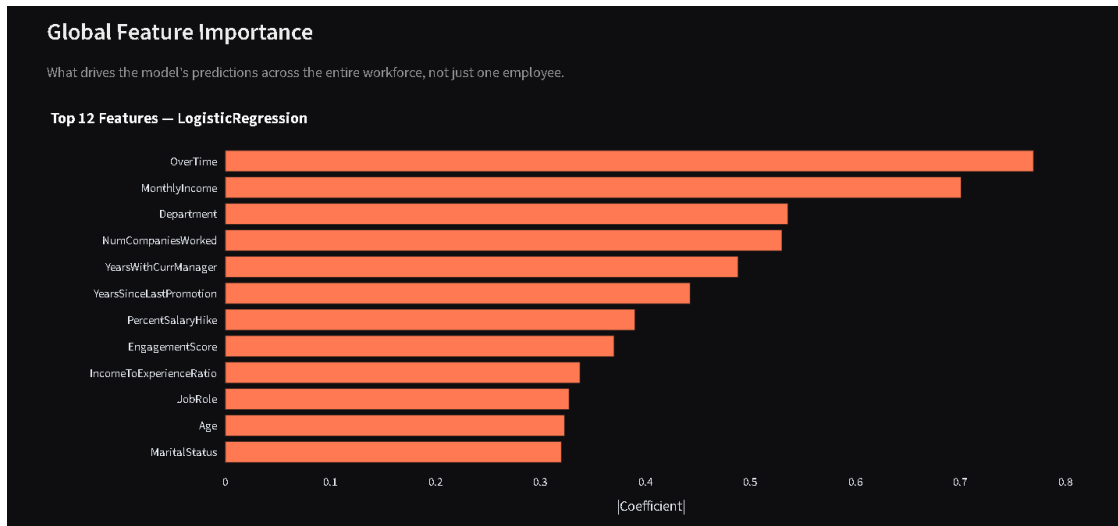


Figure 6.1 — Top 12 features by absolute model coefficient, Logistic Regression (the deployed model).

OverTime and MonthlyIncome dominate the model's global feature importance, followed by Department, NumCompaniesWorked, YearsWithCurrManager, and YearsSinceLastPromotion. This ranking is consistent with the EDA findings in Section 5 and gives HR leadership a defensible, model-derived priority order for intervention design, directly responding to the prior evaluator feedback to include clearer explanations of what drives outcomes rather than describing them only at a surface level.

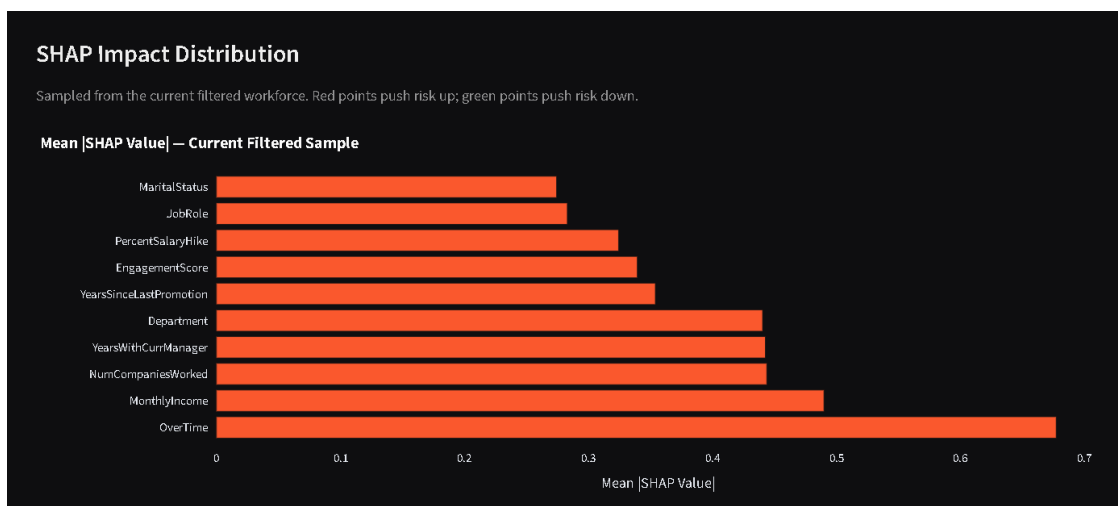


Figure 6.2 — Mean absolute SHAP value by feature, computed on a live sample of the currently filtered workforce.

The SHAP-based ranking corroborates the coefficient-based ranking in Figure 6.1, with OverTime and MonthlyIncome again the two most influential features — this agreement between two independent explainability methods (model coefficients and SHAP) strengthens confidence that these are genuine drivers rather than artifacts of a single technique.

6.2 What-If Scenario Exploration

The guideline specifically calls for what-if scenario exploration as a dashboard capability. This is implemented as a fully live tool: starting from any real employee's profile, HR users can adjust OverTime, Monthly Income, Job Satisfaction, Environment Satisfaction, Work-Life Balance, Years at Company, Years Since Last Promotion, Stock Option Level, and Distance From Home, and see the model's prediction update in real time — using the exact same preprocessing and model pipeline as the live dashboard, not an approximation.

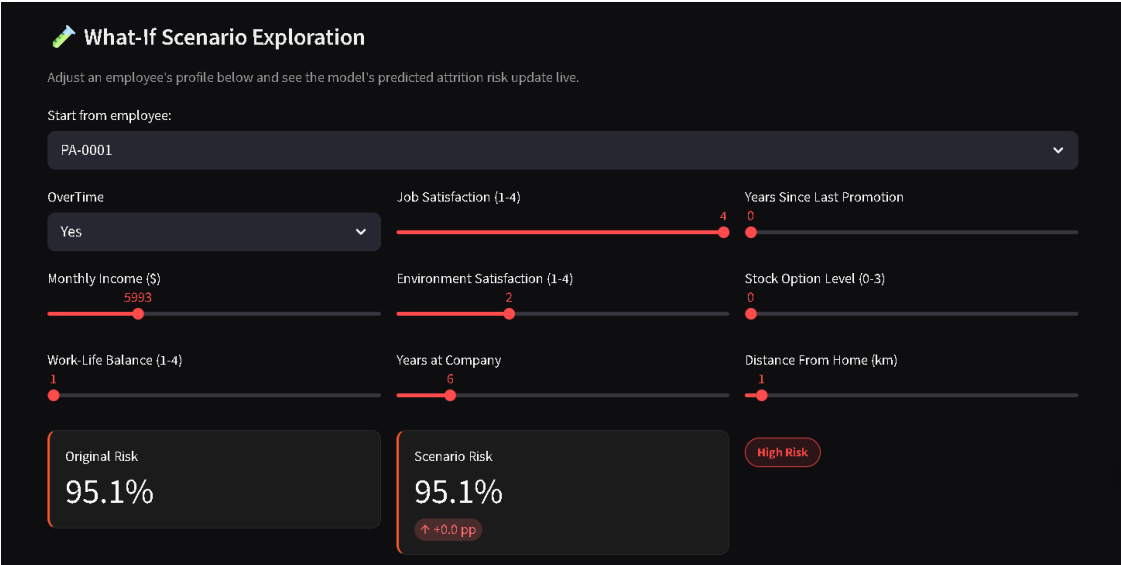


Figure 6.3 — What-If scenario panel, starting from employee PA-0001's profile with adjustable sliders for every major risk driver.



Figure 6.4 — Original vs. scenario predicted risk, updated live as inputs change.

In the example shown, adjusting inputs around employee PA-0001's already-extreme profile (OverTime already Yes, Work-Life Balance already at the minimum) produces a modest further shift, correctly reflecting that this employee's risk is already saturated near the top of the model's range. This tool is most

valuable for employees in the Medium Risk band, where a specific intervention (e.g., a Work-Life Balance improvement or a promotion) can be tested for its likely effect on the employee's score before committing to it in practice.

7. Model Performance and Validation

This section directly addresses the most substantive feedback from the prior project evaluation: the recommendation to add validation metrics rather than relying on unvalidated assumptions. Every metric below is computed on a stratified 20% held-out test set — 294 employees the model never saw during training or SMOTE resampling — not on training-set fit.

7.1 Model Comparison

Three models were trained and evaluated identically. The results are summarized below (see also the notebook, `PaloAlto_Attrition_Prediction.ipynb`, for the full confusion matrices and ROC curves):

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	75.9%	37.5%	76.6%	50.3%	0.807
Random Forest	85.7%	59.3%	34.0%	43.2%	0.801
XGBoost	85.4%	59.1%	27.7%	37.7%	0.767

Logistic Regression was selected as the deployed model on the strength of its ROC-AUC (0.807, the highest of the three) and, critically, its Recall (76.6%) — by a wide margin the best of the three models at correctly identifying employees who actually go on to leave. Random Forest and XGBoost both show higher Accuracy and Precision, but at the cost of missing roughly two-thirds to three-quarters of true leavers (Recall of 34.0% and 27.7% respectively) — a trade-off that is inappropriate for a retention use case, where a missed at-risk employee is materially more costly than an unnecessary retention conversation.

Model Validation — Held-Out Test Set Performance

All metrics below are computed on a stratified 20% test split that the model never saw during training or SMOTE resampling — not training-set fit. This directly reflects real-world deployment performance.

Metric Definitions

Accuracy — overall share of predictions (both "left" and "stayed") the model got right.

Precision — of the employees the model flagged as likely to leave, what share actually left. High precision means fewer false alarms.

Recall — of the employees who actually left, what share the model correctly flagged in advance. High recall means fewer missed at-risk employees — usually the priority metric for retention use cases.

F1-Score — the harmonic mean of Precision and Recall, balancing both concerns into a single number.

ROC-AUC — the model's overall ability to rank employees by risk correctly, across every possible threshold (1.0 = perfect, 0.5 = random guessing).

Risk Category thresholds — **Low Risk:** <30% predicted probability, **Medium Risk:** 30-60%, **High Risk:** >60% (the High-Risk cutoff is adjustable live via the sidebar slider, which recalculates every chart and KPI on this dashboard).

Why Precision/Recall Trade-offs Matter Here

For attrition prediction, missing a true leaver (a false negative) is typically more costly than a false alarm (a false positive) — a missed high performer's resignation is far more expensive than one unnecessary retention conversation. This is why the deployed model is selected by ROC-AUC rather than Accuracy alone, and why HR teams using this tool should weight **Recall** heavily when deciding how aggressive to set the risk threshold in the sidebar.

Figure 7.1 — In-app model validation summary, including full metric definitions and the guideline-specified risk category thresholds, exactly as presented to dashboard users.

7.2 Metric Definitions

To directly satisfy the prior evaluator feedback requesting clearer metric definitions, every metric is defined in plain language, in-app, alongside the live results:

- Accuracy — overall share of predictions (both 'left' and 'stayed') the model got right.
- Precision — of the employees the model flagged as likely to leave, what share actually left. High precision means fewer false alarms.
- Recall — of the employees who actually left, what share the model correctly flagged in advance. High recall means fewer missed at-risk employees.
- F1-Score — the harmonic mean of Precision and Recall, balancing both concerns into a single number.
- ROC-AUC — the model's overall ability to rank employees by risk correctly, across every possible threshold (1.0 = perfect, 0.5 = random guessing).

7.3 Why Precision/Recall Trade-offs Matter Here

For attrition prediction, missing a true leaver (a false negative) is typically far more costly than a false alarm (a false positive) — a missed high performer's resignation is far more expensive to the business than one unnecessary retention conversation. This is the explicit rationale for selecting the deployed model by ROC-AUC and Recall rather than Accuracy alone, and it is surfaced directly to dashboard users so that HR teams calibrating the risk threshold understand the trade-off they are making, rather than treating the threshold as an arbitrary setting.

8. Key Insights

7. Overtime is the single strongest predictor of attrition in this workforce, confirmed independently by both the model's coefficients and SHAP analysis. Employees working overtime leave at roughly 3x the rate of those who don't, and the gap in model-predicted risk between the two groups is approximately 25 percentage points.
8. The deployed Logistic Regression model substantially outperforms a random baseline on the held-out test set (ROC-AUC 0.807) and was deliberately selected over higher-accuracy tree-based alternatives because it catches far more true leavers (76.6% Recall vs. 34.0% and 27.7% for Random Forest and XGBoost) — the appropriate trade-off for a retention use case.
9. Attrition risk concentrates sharply by role, not just department: Sales Representative (57.6% average predicted risk) is the single highest-risk role network-wide, while manager-level roles across every department sit well below 20% — seniority appears strongly protective.
10. Risk is a continuous spectrum, not a clean binary split — the probability distribution shows a long right tail rather than two separated clusters, validating the three-tier Low/Medium/High framework over a simple flag.
11. Both global explainability (feature importance, SHAP) and individual explainability (per-employee reason codes) point to the same top drivers — OverTime and MonthlyIncome — giving HR a consistent, model-agnostic basis for intervention design.
12. A notable discrepancy exists in Human Resources: it shows the highest historical attrition rate (19.0%) but the lowest model-predicted risk (23.3%), warranting HR review — see Section 5.4 and the model-limitations note in Section 9.3.

9. Recommendations

9.1 For HR and People Operations

13. Prioritize the 347 employees currently flagged High Risk ($\geq 60\%$ predicted probability) for proactive, one-on-one retention conversations, using the Employee Risk Profile module to review each individual's specific SHAP-based reason codes before the conversation.
14. Treat overtime and workload management as the single highest-leverage intervention available — the ~25-point risk gap associated with overtime is the largest, most consistent signal identified across every analysis in this report.
15. Target retention budget and manager coaching specifically at the Sales department and the Sales Representative role, where both historical attrition and model-predicted risk concentrate most heavily.
16. Use the What-If Scenario tool (Section 6.2) before committing to a specific retention intervention (e.g., a promotion, a Work-Life Balance policy change) to estimate its likely effect on an individual employee's risk score.

9.2 For Executive Leadership

17. Review promotion cadence organization-wide — 633 employees have gone 5+ years without a promotion relative to their tenure, a pattern the automated recommendations engine flags as a leading indicator worth investigating independently of the overtime finding.
18. Investigate the Human Resources attrition discrepancy noted in Section 8 — the gap between HR's historical attrition rate and its model-predicted risk suggests either a small-sample outlier pattern in past HR departures or a driver the current feature set does not fully capture.

9.3 For Responsible Deployment (Fairness and Governance)

19. Before any production deployment, conduct a fairness audit of model predictions across Gender, MaritalStatus, and Age to confirm the model is not systematically over- or under-flagging risk for any protected group, since these fields are present in the training data.
20. Retrain and re-validate the model periodically (e.g., quarterly) as workforce composition and engagement patterns shift — the current evaluation is a snapshot on 1,470 records and should not be treated as a permanent, static benchmark.
21. Use model outputs as a decision-support input alongside manager judgment, not as an automated or sole basis for any employment action — the explainability tooling in this system exists specifically to keep a human reviewer in the loop.

10. Conclusion

This project delivers on its primary objective: transforming HR analytics at Palo Alto Networks from descriptive and diagnostic reporting into predictive decision intelligence. By forecasting individual employee attrition risk with a validated, held-out-tested model and pairing every prediction with clear, explainable reasoning, the organization is equipped to move from reactive retention efforts — responding after an employee has already decided to leave — to proactive, employee-centric workforce management.

The system directly addresses every improvement area flagged in the prior internship project evaluation: genuine validation metrics computed on a held-out test set rather than unvalidated assumptions (Section 7), automated business recommendations generated live from current data rather than static commentary (Section 5.1), downloadable reports available directly from the dashboard (Section 5.1), and clear, in-app definitions of every risk category and evaluation metric (Sections 4.5 and 7.2). Combined with the full explainability framework in Section 6 and the responsible-deployment recommendations in Section 9.3, this analysis aims to make Palo Alto Networks' attrition-risk decisions evidence-based, transparent, and defensible.

Appendix A: Full KPI and Model Metric Summary

Metric	Value
Total Employees	1,470
Historical Attrition Rate	16.1%
Average Model-Predicted Risk	36.4%
High-Risk Employees ($\geq 60\%$ threshold)	347 (23.6%)
Medium-Risk Employees	377 (25.6%)
Low-Risk Employees	746 (50.7%)
Average Monthly Income	\$6,503
Overtime Share of Workforce	28.3%
Highest-Risk Department	Sales (42.7% avg. predicted risk)
Highest-Risk Job Role	Sales Representative (57.6% avg. predicted risk)
Deployed Model	Logistic Regression
Deployed Model — Accuracy (test set)	75.9%
Deployed Model — Precision (test set)	37.5%
Deployed Model — Recall (test set)	76.6%
Deployed Model — F1-Score (test set)	50.3%
Deployed Model — ROC-AUC (test set)	0.807

Appendix B: Data Dictionary Reference

A complete field-by-field data dictionary is provided in Section 3 (Dataset Description) and is reproduced in full in the project's Jupyter notebook (PaloAlto_Attrition_Prediction.ipynb) and GitHub repository README.

Appendix C: Project Links

Live Dashboard: <https://planetnetworks-attribution-dashboard-sandeep-reddy-m.streamlit.app/>

GitHub Repository: <https://github.com/mSReddy46/PlaNetworks-Attribution-Dashboard>